



Rapid fringe decoupling for transparent object profiling using global optimization in modal wavefront reconstruction

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ABSTRACT

Phase measurement deflectometry is widely employed to reconstruct surface profiles by capturing modulated fringe patterns reflected off objects, especially for high-reflective surfaces. However, transparent objects like lenses, screens, and glass pose challenges due to overlapping fringes caused by reflections from both front and back surfaces, leading to difficulties in phase decoupling. Conventional methods relying on pixel-wise nonlinear optimization (Pw-NLP) suffer from poor robustness, high dependency on initial conditions, and inefficiency. In this paper, we propose a globally optimized approach based on modal wavefront reconstruction (MWR-GO). This method enables rapid decoupling of fringes from both the upper and lower surfaces of transparent objects in a single, efficient iteration. Our approach enhances the speed, precision, and robustness of three-dimensional reconstruction for transparent objects' surfaces.

1. Introduction

Transparent objects have extensive applications across a range of fields including optical components, virtual reality equipment, eyewear, automotive head-up displays (HUD) and windshields, as well as protective glass for computer, communication, and consumer electronics (3C) devices. The precision of their three-dimensional form is crucial for system accuracy, image clarity, durability, and stiffness. Therefore, assessing the surface morphology during the manufacturing process is vital to enhance production standards.

Deflectometry has developed rapidly in recent years, characterized by its simplicity, wide dynamic range, high speed, and high precision, and it can easily achieve in-situ measurements of any arbitrary freeform mirror. The Phase Measuring Deflectometry (PMD) system operates based on Snell's reflection law, generally comprising a display (typically an LCD), a camera, and the test surface. The structured light pattern displayed on the LCD reflects off the test surface and is then captured by the camera. By matching the corresponding screen and camera pixels, the test surface gradient is derived via ray tracing, followed by integration techniques to reconstruct the three-dimensional shape of any freeform surface.

Although phase measurement deflectometry can also reconstruct transparent objects with mirror-like reflective properties, it encounters issues with overlapping stripes from both the front and back surfaces.

Light from the display is reflected by both surfaces, leading to overlapping stripes captured by the camera. The additive nature of the stripe intensities significantly hinders pixel matching, thus preventing the separate reconstruction of the front and back surfaces.

Currently, the solutions to the problem of stripe overlapping are mainly divided into three categories. Firstly, the backward reflections are eliminated by using additional equipment. Drouet et al. separated the fringes of the front and rear surfaces by adding a polarizer in front of the camera [1]. D. Sprenger et al. utilized the characteristic that most minerals and organic glass are opaque to radiation below 330 nm. By adding an ultraviolet light source and a slit, an ultraviolet deflectometry method was constructed [2]. However, these methods increase the system's complexity and are not suitable for in-situ field measurements.

Secondly, Fourier Transform techniques that encompass Time Fourier Transform and Spatial Fourier Transform methods are employed. Zheng et al. successfully distinguished the front and back surfaces of transparent objects by analyzing the modulation differences in stripe frequencies over time [3]. Furthermore, Sotero et al. developed a spatial Fourier Transform method aimed at eliminating "ghost reflections" from the back surface, but this method requires specific deflectometry systems for different test components to ensure that the front and back surfaces have opposite phase flows, which lacks universality [4].

Thirdly, nonlinear least squares optimization is applied to jointly

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solve phases from both surfaces. Huang [5] and Ye [6] proposed dual-frequency phase-shifting deflectometry and multi-frequency deflectometry, respectively. Grating stripes of different frequencies and phase shifts are displayed, thereby establishing a set of equations for the phase and modulation degree in relation to intensity for both front and back surfaces. Subsequently, nonlinear least squares optimization is used to solve the surfaces to be tested pixel by pixel, thus restoring the phases of both surfaces. These approaches provide high accuracy for initial phase estimation through the use of a known model-based reverse light path tracing method. However, in most practical applications, the model is unknown, rendering this method inapplicable. To enable the application in scenarios where the model is unknown, we modify the phase initialization method of the approaches in [5] and [6] by adopting a spectrum filtering technique that does not require prior model knowledge. Since this method still retains the pixel-by-pixel phase iteration characteristic of the original methods, we refer to it as the pixel-wise nonlinear optimization method (Pw-NLP).

However, this Pw-NLP method encounters several challenges, such as phase ambiguity, phase order jumping, and low demodulation efficiency, as mentioned in Section 2. These three issues result in poor phase reconstruction accuracy for the front and rear surfaces, slow low robustness, and low processing speed, which in turn affect the accuracy of the surface reconstruction for both the front and rear surfaces.

Consequently, we propose a global iterative optimization method that employs the modal wavefront reconstruction method (MWR-GO) to construct the phases and modulation degrees of both front and back surfaces. Simulations and experiments have shown that this method simultaneously addresses phase ambiguity, phase order discontinuity, and low demodulation efficiency.

2. Modeling

2.1. Establishing and analyzing the fringe patterns superposition model

A standard phase measuring deflectometry system mainly includes a camera, tested specular objects/mirrors, and a display screen, as shown in Fig. 1. The screen shows sinusoidal fringe patterns with phase shifts. For transparent objects, such as eyeglass lenses and optical lenses, reflections occur on both the front and back surfaces. A single camera pixel, C, corresponds to two distinct pixels on the display screen, S_1 and S_2 , as depicted in Fig. 1. Therefore, different phases for the front and back surfaces are simultaneously captured by the camera, leading to the superposition of stripe intensities. $I_n(x, y)$ is defined to represent the distribution of light intensity captured by the camera during the n th phase shift, as shown in Eq. (1).

$$I_n(x, y) = A(x, y) + B_f(x, y)\cos(k\varphi_f(x, y) + \delta_n) + B_r(x, y)\cos(k\varphi_r(x, y) + \delta_n) \quad (1)$$

$A(x, y)$ represents the ambient background light, $B_f(x, y)$ and $B_r(x, y)$

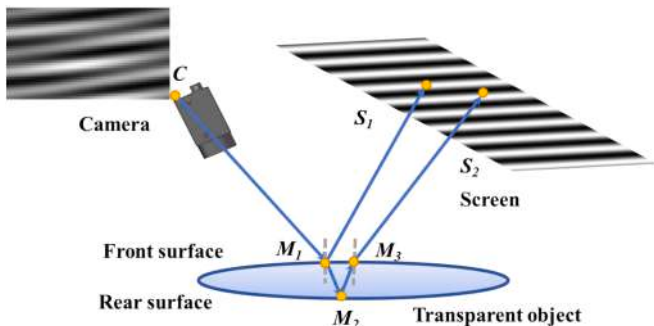


Fig. 1. Fringe Moiré Produced by a Dual-Layer Transparent Object.

denote the modulation intensities of the front and rear surfaces, respectively, while φ_f and φ_r represent their respective phases. $\delta_n = 2\pi n/N$ ($n = 0, 1, 2, \dots, N-1$) denotes the phase shift of the sinusoidal fringe patterns, where N represents the total number of phase shifts. k represents the ratio between frequencies in the case of multi-frequency fringes.

To demodulate the phase and modulation of the front and rear surfaces, multiple sets of fringe patterns with phase shifts or frequencies are generated, forming a series of equations. Eq. (1) is typically transformed into the form of Eq. (2), where the ambient light component is eliminated. The left-hand side represents a linear combination of patterns' light intensity, while the right-hand side consists of nonlinear combinations of the modulation and phase of the front surface and the modulation and phase of the rear surface. For example, the dual-frequency phase-shifting method, as shown in Eq. (3), involves phase shifts with $N = 4$ and $k = k_0, k_1$ [5]. Similarly, the multi-frequency phase-shifting method, shown in Eq. (4), uses $k = k_0, k_1, k_2, k_3$, and k_4 to construct a set of equations for the retrieval of phase and modulation [6]. The core equation sets for optimization are shown in Fig. 2(a) and (b).

Regardless of the core equation set used, traditional methods (also called the Pw-NLP method) typically follow a pixel-by-pixel approach, as illustrated in Fig. 4(a). Nonlinear optimization (NLP) is first applied to an individual pixel, and the result serves as the initial guess for neighboring pixels in an iterative process. A flood-fill algorithm is then employed to obtain a global solution.

$$\sum_{i=1}^n a_i I_i = g(B_f, \varphi_f) + h(B_r, \varphi_r) \quad (2)$$

$$\frac{2}{N} \sum_{n=0}^{N-1} I_n(x, y) \cdot \cos \delta_n = B_f(x, y) \cos(k\varphi_f(x, y)) + B_r(x, y) \cos(k\varphi_r(x, y)) \quad (3)$$

$$I_w - I_0 = [\cos((k_w - 1)\varphi_f) - 1] \cos \varphi_f B_f - \sin((k_w - 1)\varphi_f) \sin \varphi_f B_f + [\cos((k_w - 1)\varphi_r) - 1] \cos \varphi_r B_r - \sin((k_w - 1)\varphi_r) \sin \varphi_r B_r \quad (4)$$

However, the Pw-NLP method still faces the main issues identified through our simulations and experiments. (1): Phase ambiguity. In the aforementioned equations, the phase of the front surface φ_f and the phase of the back surface φ_r are equivalent in the formula, which leads to phase confusion especially when the phases of the front and back surfaces are close, as depicted in Fig. 2(c). (2): Phase order jumping. Experiments have shown that the final optimization accuracy heavily depends on the initial phase values, which can lead to order jumping and susceptibility to noise, as illustrated in Fig. 2(d). This is because the relationship between phase and intensity is periodic with a 2π cycle, resulting in countless local minima throughout the solution space. Fig. 3(a) displays the residual distributions for phase calculation. The 3D plot shows the residual values for different φ_f and φ_r values. The red dots represent the global minimum values, while the yellow dots represent the local minimum values. Fig. 3(b) shows the path of phase iteration. Different initial values will lead to different iteration paths and convergence points. If the initial phase value significantly deviates from the true value, the optimization process will converge to a local minimum rather than the global minimum. (3): Low demodulation efficiency. Fig. 2(e) lists the time consumption for different image sizes (64×64 pixels and 2000×3000 pixels), which are 5 s and 122 min, respectively. The Pw-NLP method results in low efficiency of phase calculation, which is not conducive to in-situ measurements.

2.2. Global optimization model based on modal wavefront reconstruction

We analyzed the three issues, phase ambiguity, phase order jumping, and low demodulation efficiency, which collectively highlight a fundamental challenge. In practice, the front and back surfaces of transparent

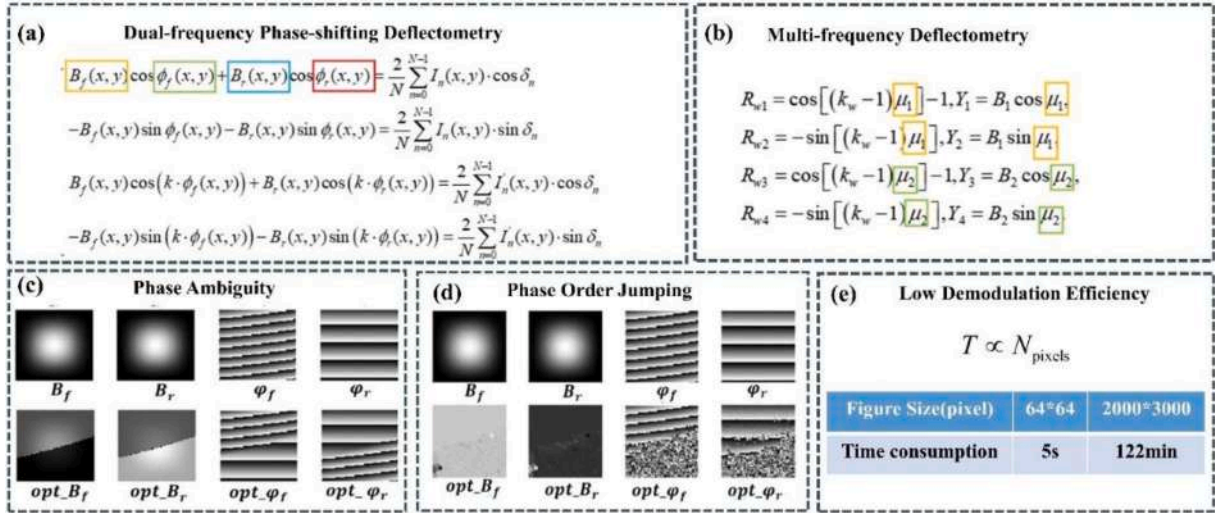


Fig. 2. The Pw-NLP method and its three challenges. (a) Dual-frequency phase-shifting deflectometry. (b) Multi-frequency deflectometry. (c) Phase ambiguity. (d) Phase order jumping. (e) Low demodulation efficiency.

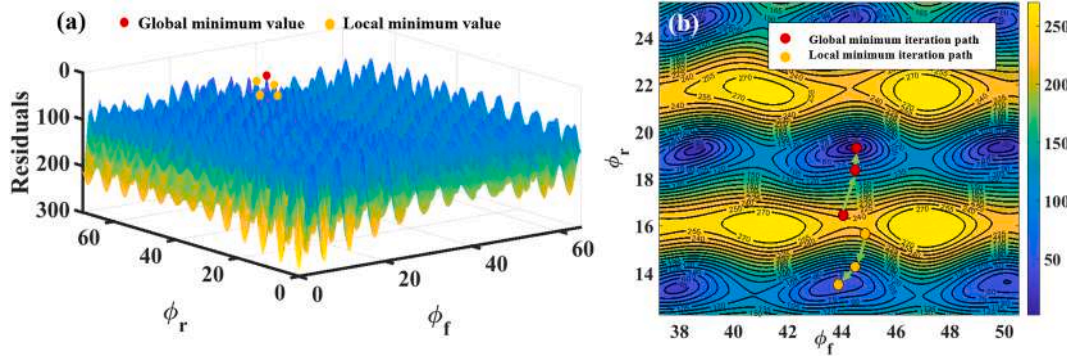


Fig. 3. Residual distributions for phase calculation, iteration paths, and convergence points for different initial values. (a) Residual distributions for phase calculation. (b) Path of phase iteration.

objects exhibit continuous behavior. Thus, the associated phases ϕ_f and ϕ_r are also continuous. However, the Pw-NLP method neglects the continuity of phase between adjacent pixels.

In conclusion, these issues arise because the method fails to adequately leverage the inherent physical properties of the surface in real-world conditions. Therefore, solving for these surface phases from a

global perspective is warranted.

Given the continuity of the four unknown quantities in the x and y directions, employing modal wavefront reconstruction [7,8] is feasible. We propose a global optimization method based on modal wavefront reconstruction (MWR-GO), as shown in Fig. 4(b). This technique utilizes a series of mutually orthogonal polynomials to model the modulation

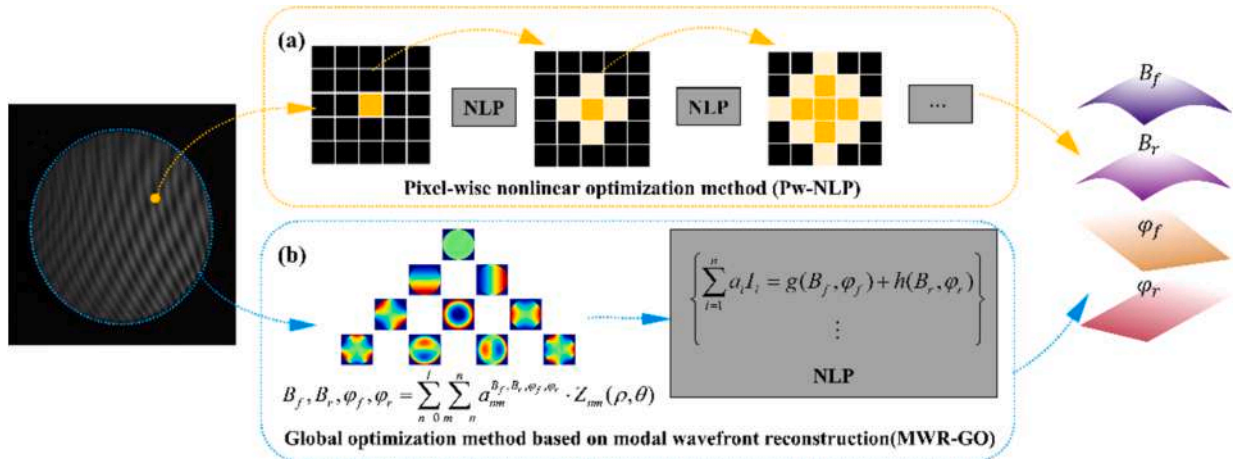


Fig. 4. Comparison of two different phase decoupling methods. (a): Pixel-wise nonlinear optimization method (Pw-NLP). (b): Global optimization method based on modal wavefront reconstruction (MWR-GO).

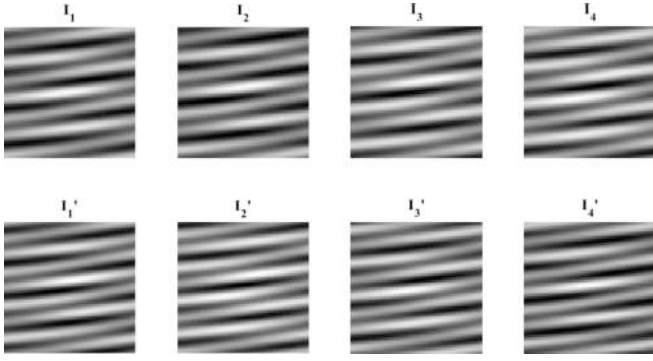


Fig. 5. Four-step phase-shifted fringe patterns with two frequencies I_n and I_n' ($n = 1, 2, 3, 4$).

intensities and phases of the front and back surfaces. Typically, the choice of polynomials—such as Legendre, Chebyshev, or Zernike polynomials—depends on the specific surface shape being analyzed. Zernike polynomials are commonly employed for circular regions, whereas Legendre polynomials are used for rectangular areas. The modal wavefront reconstruction utilizes relatively few parameters representing the entire surface. This allows for optimizing the phase and modulation of the entire surface in an NLP process, significantly eliminating the need for pixel-by-pixel calculation of modulation intensity and phase. With a single iteration, the phase decoupling of the entire front and rear surfaces can be achieved, ensuring phase continuity and improving the robustness of phase retrieval, while avoiding phase ambiguity and phase order jumping.

For instance, using Zernike polynomials, the four unknowns B_f , B_r , φ_f , and φ_r are represented as linear combinations of Zernike polynomial terms $Z_{nm}(\rho, \theta)$, where n ranges from 0 up to 1, and m varies from $-n$ to n in steps of 2. The coefficients $[a_{nm}^{Bf}, a_{nm}^{Br}, a_{nm}^{\varphi f}, a_{nm}^{\varphi r}]$ represent the fitting parameters for these polynomials. We choose an appropriate polynomial order l based on the complexity of the front and rear surface contours. Here, l is set to 5, covering the first 21 terms for a_{nm}^{Bf} , with similar operations for the other parameters, as shown in Eq. (5). Similarly, B_f , B_r , φ_f , and φ_r can also be expressed as a linear combination of the product of Legendre polynomials $D(x)$ and $D(y)$ along the x and y directions. Here, k represents the order of the Legendre polynomials, as also shown in Eq. (5). Zernike polynomials are chosen for circular regions due to their efficiency in representing smooth, rotationally symmetric surface variations, while Legendre polynomials are selected for rectangular areas because of their orthogonality and suitability for describing variations along both x and y directions. In the following process, we will consistently use Zernike fitting for modeling.

$$\left\{ \begin{array}{l} B_f(x, y) = \sum_{n=0}^l \sum_{m=-n}^n a_{nm}^{Bf} \cdot Z_{nm}(\rho, \theta) \\ B_r(x, y) = \sum_{n=0}^l \sum_{m=-n}^n a_{nm}^{Br} \cdot Z_{nm}(\rho, \theta) \\ \varphi_f(x, y) = \sum_{n=0}^l \sum_{m=-n}^n a_{nm}^{\varphi f} \cdot Z_{nm}(\rho, \theta) \\ \varphi_r(x, y) = \sum_{n=0}^l \sum_{m=-n}^n a_{nm}^{\varphi r} \cdot Z_{nm}(\rho, \theta) \end{array} \right. \quad \text{or} \quad \left\{ \begin{array}{l} B_f(x, y) = \sum_{m=0}^k \sum_{n=0}^k a_{mn}^{Bf} \cdot D_m(x) D_n(y) \\ B_r(x, y) = \sum_{m=0}^k \sum_{n=0}^k a_{mn}^{Br} \cdot D_m(x) D_n(y) \\ \varphi_f(x, y) = \sum_{m=0}^k \sum_{n=0}^k a_{mn}^{\varphi f} \cdot D_m(x) D_n(y) \\ \varphi_r(x, y) = \sum_{m=0}^k \sum_{n=0}^k a_{mn}^{\varphi r} \cdot D_m(x) D_n(y) \end{array} \right. \quad (5)$$

The MWR-GO method is shown in Eq. (6). Using the dual-frequency phase-shifting method as an example, the objective function is defined as the sum of the squares of the residuals f_1, f_2, f_3 , and f_4 , which represent the differences between the modeled values (left side of the equation) and observed values (right side of the equation) in Fig. 2(a). The parameters need to be optimized, denoted by J , consist of the Zernike

fitting parameters $[a_{nm}^{Bf}, a_{nm}^{Br}, a_{nm}^{\varphi f}, a_{nm}^{\varphi r}]$. The optimal set of parameters by applying nonlinear least squares optimization [9,10], denoted by J^* , represents the best-fit values for these Zernike coefficients.

$$\left\{ \begin{array}{l} J^* = \operatorname{argmin} \left\{ \sum_{i=1}^4 [f_i(J)]^2 \right\} \\ J = [a_{nm}^{Bf}, a_{nm}^{Br}, a_{nm}^{\varphi f}, a_{nm}^{\varphi r}] \\ f_1 = B_f(x, y) \cos \varphi_f(x, y) + B_r(x, y) \cos \varphi_r(x, y) - \frac{2}{N} \sum_{n=0}^{N-1} I_n(x, y) \cdot \cos \delta_n \\ f_2 = -B_f(x, y) \sin \varphi_f(x, y) - B_r(x, y) \sin \varphi_r(x, y) - \frac{2}{N} \sum_{n=0}^{N-1} I_n(x, y) \cdot \sin \delta_n \\ f_3 = B_f(x, y) \cos(k \cdot \varphi_f(x, y)) + B_r(x, y) \cos(k \cdot \varphi_r(x, y)) - \frac{2}{N} \sum_{n=0}^{N-1} I'_n(x, y) \cdot \cos \delta_n \\ f_4 = -B_f(x, y) \sin(k \cdot \varphi_f(x, y)) - B_r(x, y) \sin(k \cdot \varphi_r(x, y)) - \frac{2}{N} \sum_{n=0}^{N-1} I'_n(x, y) \cdot \sin \delta_n \end{array} \right. \quad (6)$$

This approach effectively minimizes discrepancies between our predictive model and the actual measurements by considering the entire surface as a whole. This holistic perspective not only enhances the accuracy of our model but also addresses fundamental issues such as local phase ambiguity and order jumping. Additionally, this method significantly improves the efficiency of the measurement process, offering a more streamlined and reliable means of phase acquisition.

3. Simulations

3.1. Simulation model

To further validate the reliability of the aforementioned method, we conducted a series of simulation experiments. First, we defined a 64×64 image space and generated a Hann window function to simulate the spatial non-uniformity of ambient light $A(x, y)$ and the modulation of the front and rear surfaces. Specifically, the functions are defined as follows: $A(x, y) = 121 + 14 \cdot \text{Hann}(x, y)$, $B_f(x, y) = 42 + 18 \cdot \text{Hann}(x, y)$, $B_r(x, y) = 24 + 8 \cdot \text{Hann}(x, y)$. The phases of the front and rear surfaces are expressed in Eq. (7) and Eq. (8), respectively. The ground truths of modulation and phase of the front and rear surfaces are shown in Fig. 6 (a)–(d).

$$\varphi_f = \frac{2\pi(0.1x + 5 \times 10^{-4}x^2 + y)}{9} \quad (7)$$

$$\varphi_r = \frac{2\pi y}{16} + 12 \quad (8)$$

The fringe frequencies were set to $k_0 = 1$ and $k_1 = 1.1$. This frequency setting ensures a sufficient unambiguous measurement range while enhancing the noise resistance of phase demodulation. The number of phase shifts $N = 4$. Four-step phase-shifted fringe patterns with two frequencies I_n and I_n' ($n = 1, 2, 3, 4$) can be generated. To simulate noise conditions in the experiment, we added Gaussian random noise $I_{\text{noise}} \sim N(0, \sigma_A^2)$ to the I_n and I_n' images. σ_A represents the noise of type A, the level of image noise. The generated phase-shifted images containing noise, and Fig. 5 below, respectively.

The traditional point-wise nonlinear optimization method (Pw-NLP) is applied to decouple modulation and phase, and the results are shown in Fig. 6(e)–(h). Due to image noise, the error for a single pixel will further propagate to surrounding pixels, causing the instability of the initial phase value. The 2π periodicity of the phase and intensity creates countless local minima throughout the solution space. When the instability of the initial phase value exceeds 2π , it causes cycle slips during NLP, leading to larger areas of phase errors.

To address the phase retrieval failures and inefficiency issues that the Pw-NLP method cannot resolve, we apply the global optimization method based on modal wavefront reconstruction (MWR-GO). We devised a scheme to generate initial values for the iteration to verify the robustness of the MWR-GO method. Firstly, according to Eq. (5), Zernike polynomials were used to reconstruct $B_f(x, y)$, $B_r(x, y)$, $\phi_f(x, y)$, and $\phi_r(x, y)$, obtaining $J_{truth} = [a_{nm}^{Bf}, a_{nm}^{Br}, a_{nm}^{\phi f}, a_{nm}^{\phi r}]$. Then, noise $\varepsilon \sim N(0, \sigma_B^2)$ was added to the true values J_{truth} , as shown in Eq. (9). ε follows a normal (Gaussian) distribution with a mean of 0 and a variance of σ_B^2 . σ_B represents the noise of type B, the initial iterative noise. After noise processing, the modulation and phase of the front and rear surfaces exhibit varying degrees of distortion, including translation, tilt, defocus, and other aberrations.

$$J_{noise} = J_{truth} \times (1 + \varepsilon) \varepsilon \sim N(0, \sigma_B^2) \quad (9)$$

We used J_{noise} as the initial value for the iteration, substituting it into Eq. (6). The iteration was completed using the MWR-GO method, eventually converging to J^* . Based on Eq. (5), $B_f(x, y)$, $B_r(x, y)$, $\phi_f(x, y)$, and $\phi_r(x, y)$ are restored, as shown in Fig. 6(i)–(l). It demonstrates that the phase and modulation can be accurately recovered. The MWR-GO method demonstrates a convergence time of 1.77 s, significantly enhancing the speed of convergence compared to the Pw-NLP method, which requires 16.24 s. This improvement is expected to become even more pronounced as the image size increases.

The optimized values shown in Fig. 6(i)–(l) from the MWR-GO method were treated as three-dimensional surfaces. These were then analyzed for errors relative to the ground truth surfaces shown in Fig. 6(a)–(d). The root mean square (RMS) and peak-to-valley (PV) results are presented in Table 1. These results were obtained under the conditions of $\sigma_A = 3$, $\sigma_B = 0.03$.

Through the MWR-GO method, the RMS of the front and back surface modulation degree B_f and B_r , respectively decreased from 0.9562 (Gray Level, GL) and 1.1086 (GL) to optimized values of 0.2167 (GL) and

0.1080 (GL). The RMS of the front and back surface phases ϕ_f and ϕ_r decreased from 1.1037 (rad) and 0.9142 (rad) to 6.7712e-4 (rad) and 1.8028e-3 (rad) respectively. Since the contribution curvature κ_i of different variables in the objective function varies, as shown in Eq. (10). It is evident that $\kappa_{\phi_f} \approx \kappa_{\phi_r} \gg \kappa_{B_f} \approx \kappa_{B_r}$, therefore the convergence accuracy of phase value is much higher than that for modulation degree. In practical reconstruction, only ϕ_f and ϕ_r are utilized to determine the matching of corresponding pixels between the screen and the camera. Thus, the reconstruction accuracy of the B_f and B_r does not affect the reconstruction accuracy of the front and back surfaces directly.

$$\kappa_i = \frac{\partial^2 f}{\partial x_i^2} \quad (10)$$

To further elaborate on how the MWR-GO method stands out in terms of its approach and performance, we have provided a detailed comparison of the relationships, differences, and performance across dual-frequency Phase-shifting Deflectometry, multi-frequency Deflectometry, Pw-NLP, and MWR-GO in the Supplementary (Appendix A), highlighting the unique advantages of our method through simulation analysis.

Table 1

The PV and RMS of B_f , B_r , ϕ_f , and ϕ_r before and after MWR-GO. PV: peak-to-valley. RMS: root mean square. Initial: the initial values under the conditions of $\sigma_A = 3$, $\sigma_B = 0.03$. MWR-GO: the optimized values by MWR-GO.

		B_f	B_r	ϕ_f	ϕ_r
PV	Initial	3.5387	5.5528	3.5879	3.9946
	MWR-GO	0.7147	0.3912	2.5924e-3	5.3172e-3
RMS	Initial	0.9562	1.1086	1.1037	0.9142
	MWR-GO	0.2167	0.1080	6.7712e-4	1.8028e-3

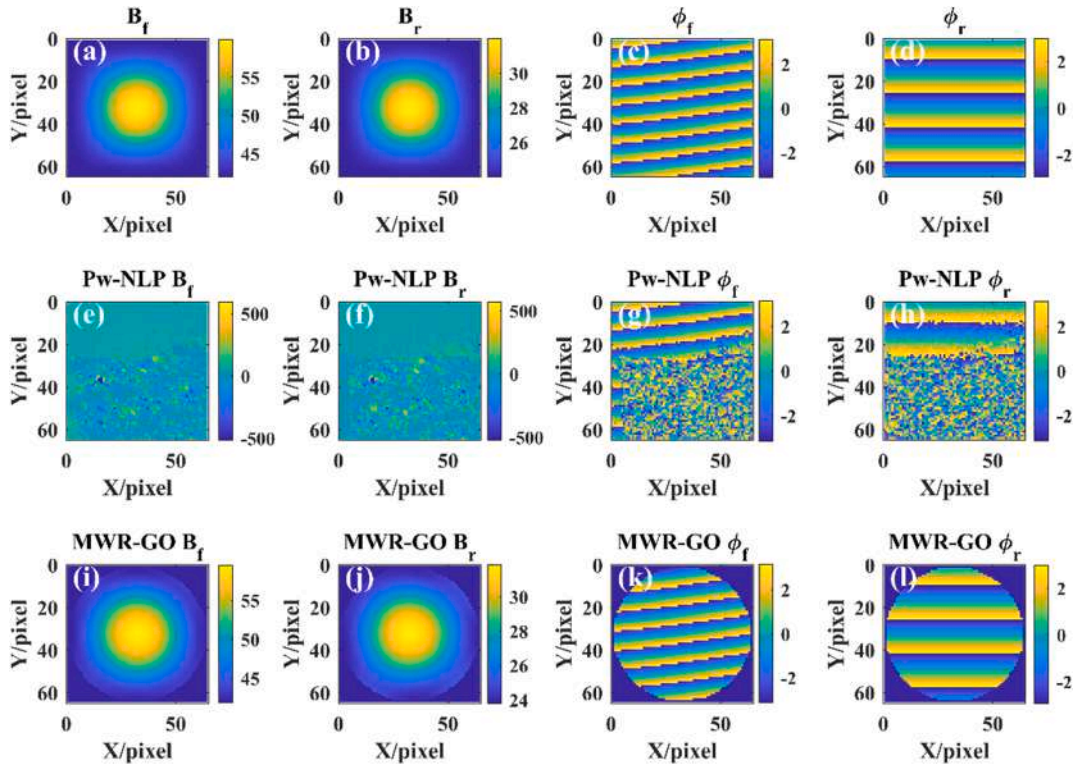


Fig. 6. 3D profiles of modulation and phase of front and rear surfaces before and after optimization. (a)–(d) The ground truth of B_f , B_r , ϕ_f , and ϕ_r (e)–(h): The optimized values of B_f , B_r , ϕ_f , and ϕ_r by point-wise nonlinear optimization method (Pw-NLP). (i)–(l): The optimized values of B_f , B_r , ϕ_f , and ϕ_r are achieved using the globally optimized modal wavefront reconstruction method (MWR-GO).

3.2. Noise analysis

To further verify the robustness and convergence of the MWR-GO method, the simulations are conducted as below. In terms of robustness, we separately tested the optimization accuracy of $B_f(x, y)$, $B_r(x, y)$, $\varphi_f(x, y)$, and $\varphi_r(x, y)$ under different image noise (σ_A) and initial iteration noise (σ_B), and the results are shown in Fig. 7(a) and Fig. 7(b), respectively. From Fig. 7(a), it can be seen that when $\sigma_A \leq 10$ (GL), the RMS and PV of the phase values of φ_f and φ_r are both less than 0.1 rad. Assuming the period of sinusoidal stripes on the display screen is 5 mm, the spatial position deviation of corresponding screen pixels is approximately 0.08 mm. Assuming the distance of the screen relative to the world coordinate system is $D = 200$ mm, the gradient deviation can be calculated as shown in Eq. (11). If the spacing of the reconstructed grid is 1 mm, the corresponding height error of the reconstruction can be estimated as 2×10^{-4} mm. It can be seen that the MWR-GO method has strong robustness in terms of image noise. Furthermore, the method also exhibits strong robustness to initial iteration noise (σ_B) in Fig. 7(b). When $\sigma_B \leq 0.045$, the RMS and PV of the φ_f and φ_r are also less than 0.1 rad. However, when $\sigma_B > 0.045$, both the surface modulation degree and phase rise sharply, indicating convergence failure when the initial iteration parameter error is too large. This behavior can be attributed to the objective function used in the MWR-GO method. As described in Eq. (6), the objective function combines the modulation term (e.g., $B_f(x, y)$) and the cosine phase factor (e.g., $\cos(\varphi_f)$), resulting in multiple possible combinations of the optimization variables. If the initial iteration parameters, such as the lower-order Zernike terms $Z_{0,0}$ (Piston), $Z_{1,-1}$ (x-tilt), and $Z_{1,1}$ (y-tilt), exhibit significant directional deviation, the optimization process may fail to converge.

To address this limitation, a hybrid approach combining spectral filtering for initial phase estimation and the MWR-GO method was adopted in Section 4. This combination leverages the high robustness of spectral filtering to low-order Zernike terms while compensating for its limited performance on high-order terms through the strong noise resistance of the global optimization method. As demonstrated in the experimental results, this hybrid approach significantly improves the overall robustness to noise, even under conditions where convergence is theoretically challenging in simulations.

$$\Delta px = \frac{1}{2} \frac{\Delta x}{D} = 2 \times 10^{-4} \quad (11)$$

Furthermore, in terms of convergence, we compared the effects of image noise (σ_A) and initial iteration noise (σ_B) on the convergence speed and accuracy of the iterations, as shown in Fig. 8. Fig. 8(a) illustrates the iteration convergence curves under different image noise ($\sigma_A = 0, 1, 2 \dots 10$, $\sigma_B = 0.03$), where the convergence accuracy decreases with increasing image noise while the convergence speed remains relatively constant. Fig. 8(b) shows the iteration convergence curves

under different initial iteration noise ($\sigma_A = 2$, $\sigma_B = 0.005, 0.010, 0.015 \dots 0.030$), where higher initial iteration noise requires more iteration steps and time to achieve convergence, but the convergence accuracy remains the same. In summary, image noise mainly affects convergence accuracy, and initial iteration noise mainly affects convergence speed.

4. Experiments

4.1. Setup

We have constructed a phase-measuring deflectometry setup as illustrated in Fig. 9. The system includes a monochrome camera MV-CS200-10GM with a resolution of 5472×3648 pixels. The camera's working distance is 250 mm, and it is equipped with a lens of 16 mm focal length, providing a field of view of $240 \text{ mm} \times 175 \text{ mm}$. The display screen used is an iPad 9 with a size of 10.2 in. When displayed at a 16:9 aspect ratio, the dimensions are 8.16 in. $\times$ 4.59 in. (207.264 mm $\times$ 116.586 mm). The display screen shows monochromatic green sinusoidal fringe patterns to reduce the impact of fringe blurring caused by internal dispersion of the medium during reflection from the rear surface of the tested lens. Shade cloth covers the experimental setup to reduce the interference of ambient light in the experiment. An adjustable tripod controls the height and orientation of the display screen to ensure that the virtual image is accurately captured by the camera. Our computing platform is equipped with an 11th Gen Intel® Core™ i7-11700KF @ 3.60 GHz, with Windows 11 as the operating system. The software used for the experiments is MATLAB (version 2021b).

To validate the effectiveness of the MWR-GO method, we employed this model to reconstruct double-layer transparent objects. Unlike simulations, the initial phase values of the front and rear surfaces are unknown and need to be determined through ray tracing under a pre-established profile [11]. However, in practical measurements, the profile and pose of the object to be measured are difficult to obtain, making this initial value acquisition method impractical for real applications. Therefore, we adopt a spatial filtering method to preliminarily separate the phases of the front and rear surfaces [12–14], followed by a dual-frequency fringe temporal phase unwrapping technique [15,16], obtaining the absolute initial phase values of the front and rear surfaces. The detailed flowchart is shown in Fig. 10.

4.2. Initial iteration phase preprocessing

For double-layer transparent objects (such as lenses), the images captured by the camera are shown in Module 1. These images undergo a Fast Fourier Transform (FFT) to generate the Fourier-transformed images, and then the logarithmic spectrum of the Fourier-transformed images is calculated as $\log_{\text{spectrum}}(u, v) = \log(1 + |F_{\text{center}}(u, v)|)$ enhancing the dynamic range and visualization effect, as shown in

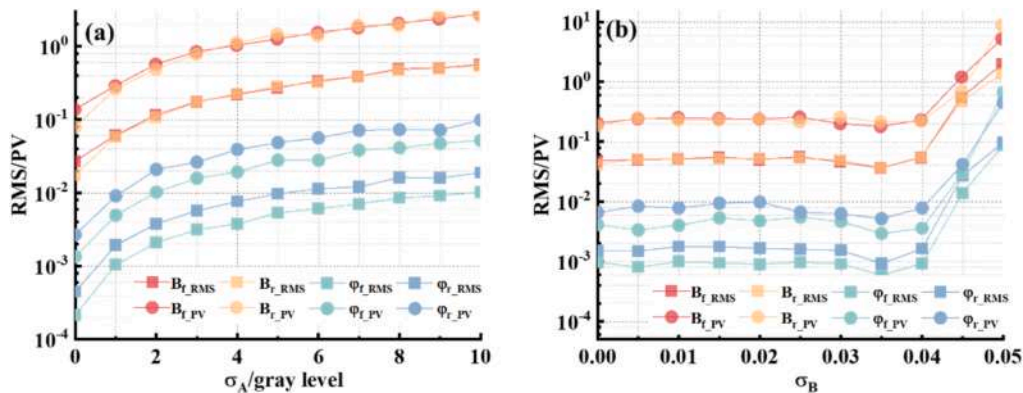


Fig. 7. The optimization accuracy (RMS and PV) of B_f , B_r , φ_f , and φ_r (a) The optimization accuracy under different image noise σ_A . (b) The optimization accuracy under different initial iteration noise σ_B .

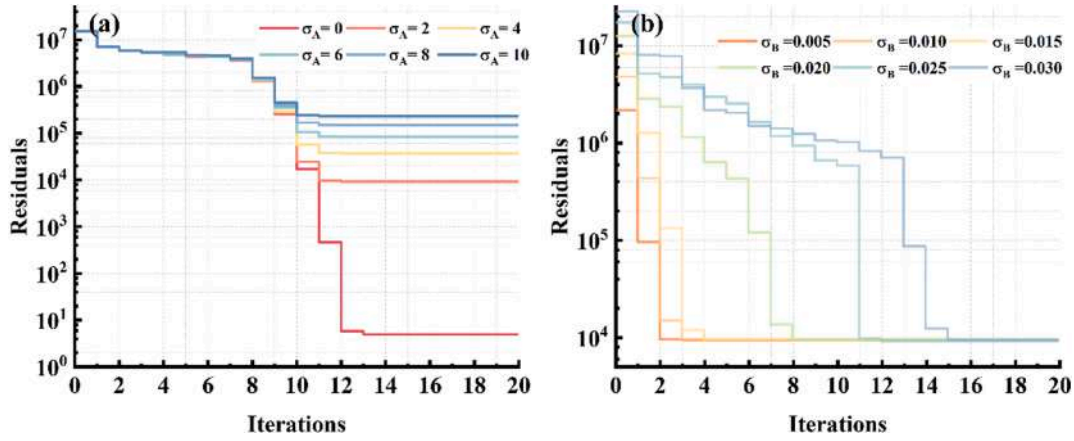


Fig. 8. The convergence speed and accuracy of the iterations. (a)Results under different image noise σ_A (b)Results under different image noise σ_B .

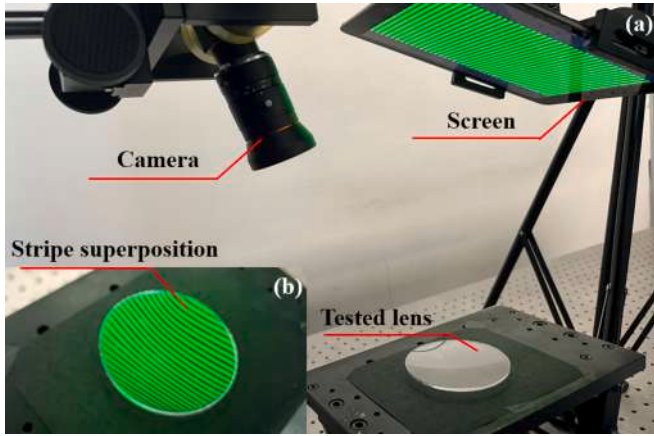


Fig. 9. The phase measuring deflectometry setup. (a) Experimental setup.(b) Stripe superposition image from the camera perspective.

Module 2. Due to the different surface profiles of the front and rear surfaces, their spatial modulation of the sinusoidal fringes varies, resulting in a central peak in the frequency domain representing ambient light, and two pairs of symmetric peaks representing the fringe patterns modulated by the front and rear surfaces. Based on the shape of the tested lens, the fringe density and direction for the front and rear surfaces can be distinguished. Each pair of symmetric peaks is used as a filter to separate the fringe information of the front and rear surfaces, as shown in Module 3.

To obtain the absolute phase of the display screen, it displays sinusoidal grating fringes with periods of $k_0 = 30$ pixels and $k_1 = 31$ pixels, respectively, each with four-step phase shifting, as shown in Module 4. Based on the single-image FFT and filtering, filters for phase-shifted fringe with the frequency of k_0 or k_1 can be calculated as the average of the four single phase-shifted image filters, as shown in Module 5. This is because phase shifting does not affect the distribution of the spectral intensity. After filtering, the truncated phase of the front and rear surfaces can be calculated separately, as shown in Module 6. We then use a dual-frequency fringe temporal phase unwrapping technique to calculate the phase orders and unwrapped phases of the front and rear surfaces, as shown in Module 7. It is important to note that the filters are Gaussian functions, and the standard deviation of the Gaussian filter is minimized under specific conditions (the mean logarithmic spectrum intensity within the range of one standard deviation is less than 0.9 times the peak logarithmic spectrum value). This dynamic adjustment ensures that the filter adapts to the data characteristics, setting appropriate standard deviations for each peak and ensuring effective filtering.

However, in practice, the peaks corresponding to the front and rear surfaces are not Gaussian but are affected by the surface specifics and the off-axis nature of the deflectometry system, resulting in directional specificity. This distorts the fringe information after Gaussian filtering, leading to phase order jumps and unwrapped phase errors, as indicated by the dashed circle areas in Module 7. To further eliminate the errors introduced by filtering and restore the phase information, the unwrapped phase with errors is used as the initial value for our MWR-GO method, as shown in Module 8. Our model iteratively minimizes the objective function $J^* = \text{argmin}(\sum_i [f_i(J)]^2)$, ensuring accurate reconstruction of the front and rear surface phases through continuous iterations.

Subsequently, based on the optimized phase information, the light rays from the same camera pixel that reflect off the front and rear surfaces can be matched with two pixels on the display screen. This allows the shape and position of the front surface to be determined through the iterative SCOTS measurement procedure [17–20]. The shape and position of the rear surface require correction of the light rays refracted into the medium. The rear surface profile converges through searching of M_3 in Fig. 1 and optimization of the ray tracing path [21].

4.3. Experimental results

We reconstructed a K9 plano-convex lens commonly used in optical experiments, with a focal length of 1000 mm. The upper surface is a spherical surface with a radius of curvature of 516.82 mm, and the material has a refractive index of 1.5163. The reconstruction region (square) is a subset of the optimized phase region (circular). The reconstruction area of the upper surface is 20 mm $\times$ 20 mm, while the reconstruction area of the lower surface is slightly smaller due to the constraints of ray tracing. Fig. 11(a) and (b) show the reconstruction results of the front and rear surfaces, and the calculated residuals with the manufacturer-provided surface profiles are shown in Fig. 12(a) and (b).

The residual RMS error for the upper surface is 778.4 nm, while the RMS error for the lower surface is 389.6 nm. It is worth noting that the low reflectivity and inhomogeneous refractive index of the glass material can affect the signal-to-noise ratio of the captured images, thereby impacting the reconstruction accuracy. The ripple characteristics of the residuals in Fig. 12(b) are due to the selection of intersection points for rays reflecting back to the upper surface M_3 , which are determined based on the discrete triangular mesh of the upper surface.

Another measurement result with an astigmatic eyeglass lens shown in Fig. 13(a) demonstrates the general applicability of this method. This type of lens has different radii of curvature and thicknesses in different directions. The shape of the lens surface can be described by an ellipsoidal paraboloid equation $x^2/a^2 + y^2/b^2 = 2z$, where x and y are along

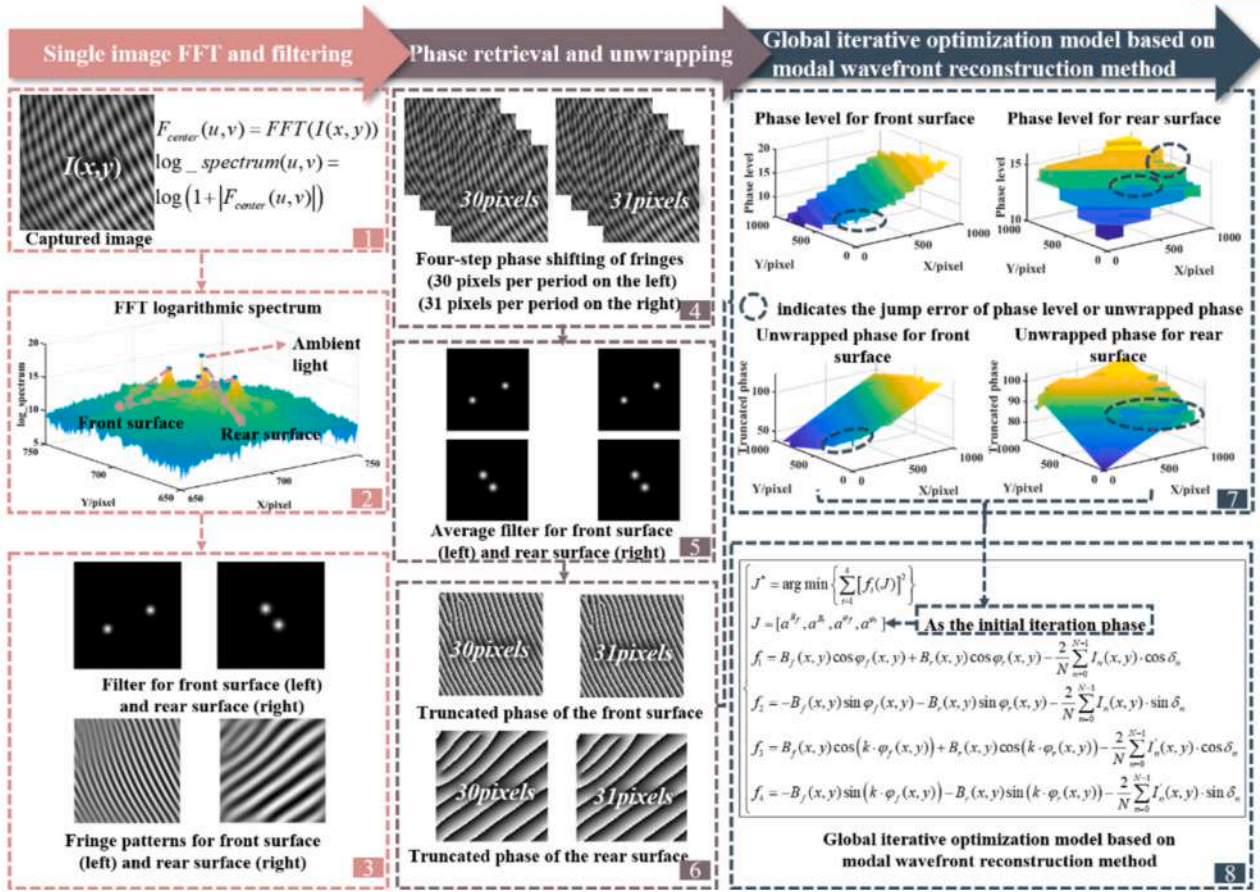


Fig. 10. Flowchart for obtaining initial phase values of front and rear surfaces.

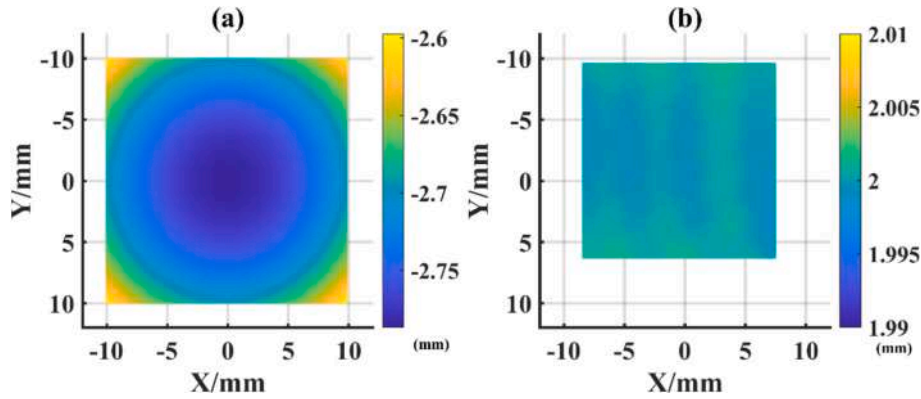


Fig. 11. The reconstruction results of a K9 plano-convex lens. (a) Front surface; (b) Rear surface.

the two principal axes, respectively. The lower surface is a spherical surface with radius R . Due to the significant curvature variations of this lens, the dynamic range of a white light interferometer is insufficient for measurement needs. Therefore, we sprayed the surface with a powder having diffuse reflective properties and then used a blue light 3D scanner (ATOS Capsule 12 M) to obtain the three-dimensional morphology of the surface with a measurement accuracy of $5 \mu\text{m}$, as shown in Fig. 13 (b). The measurement data served as the standard value. The reconstruction results of the MWR-GO method are shown in Fig. 14(a) and (b), respectively. The deviations from the standard values are shown in Fig. 15(a) and (b), respectively.

The RMS error of the upper surface is 308.5 nm , and the RMS error of the lower surface is 287.4 nm . The measurement values, standard

values, and relative deviations obtained by fitting the upper and lower surfaces with an ellipsoidal paraboloid surface and a spherical surface, respectively, are shown in Table 2. It shows that this method achieves high accuracy in extracting surface parameters of the front and rear surfaces (the major and minor axes a and b of the front surface, and the spherical radius R of the rear surface), with an accuracy within 0.1% . The experimental results indicate that this method can achieve three-dimensional reconstruction of double-layer transparent objects with arbitrary surface shapes. We have included additional experimental validation for flat and double-convex surfaces in the Supplementary (Appendix B), demonstrating the versatility and robustness of the MWR-GO method across a wider range of optical profiles.

To address the challenges of industrial in-situ applications, we

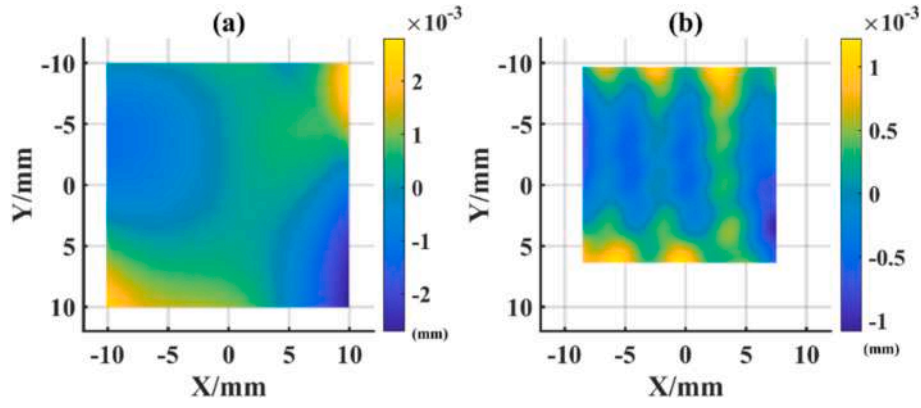


Fig. 12. The residuals of a K9 plano-convex lens. (a) Front surface; (b) Rear surface.

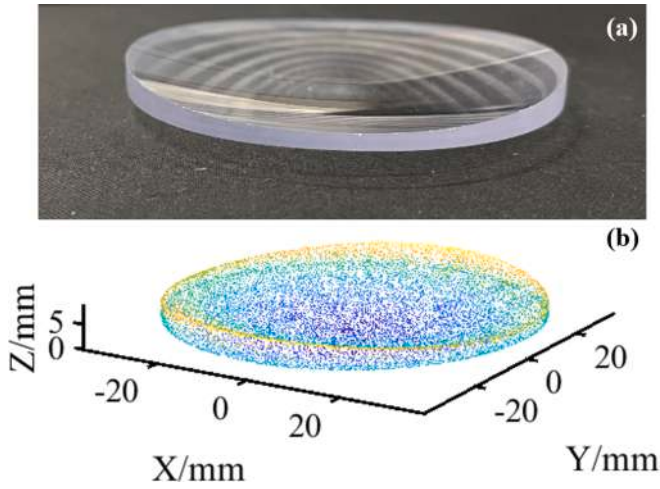


Fig. 13. An astigmatic eyeglass lens. (a) eyeglass lens captured by the camera; (b) 3D Scatter Contour Map of ATOS Capsule 12 M Scanning served as the standard value.

discuss the following aspects: environmental lighting, object positioning, real-time processing, and portability, along with proposed solutions.

1. **Environmental Lighting:** To reduce the interference of external lighting on measurement accuracy, we propose the use of a semi-enclosed test environment. Additionally, the current phase-shifting method used for phase recovery can effectively mitigate most of the interference caused by environmental lighting.

2. **Object Positioning:** For object positioning, camera-based vision positioning systems can be employed. For higher precision, fixed reference points around the object can be used to ensure more accurate positioning.
3. **Real-time Processing:** Our current system is capable of completing measurements for a single object within a few seconds. For faster or dynamic measurements, techniques such as single-frame phase demodulation are required. Some methods also incorporate deep learning algorithms for phase processing; however, a solution for single-frame phase demodulation for lens-like, double-layered objects has not yet been developed.
4. **Portability:** Our system can be mounted on mobile platforms or robotic arms for measurements. However, for more convenient, handheld devices, the main challenge is real-time object positioning, especially for reflective surfaces, which can be difficult to handle. More research is needed to adapt it for portable setups.

5. Conclusions

This paper presents a global optimization iterative algorithm based on modal wavefront reconstruction (MWR-GO), capable of decoupling the mixed fringes of double-layer transparent objects. This method fully utilizes the physical spatial continuity of the modulation and phase of the front and rear surfaces, avoiding phase ambiguities, order jumps, and low computational efficiency issues associated with per-pixel phase solving (Pw-NLP). Drawing from this, we have developed a phase deflectometry system, that can reconstruct the double-layer surface profiles of arbitrary freeform lenses. The measurement accuracy can reach the micron level, comparable to three-dimensional profilometers. Additionally, its measurement dynamic range spans from micrometers to millimeters, far exceeding that of white light interferometers, and it is

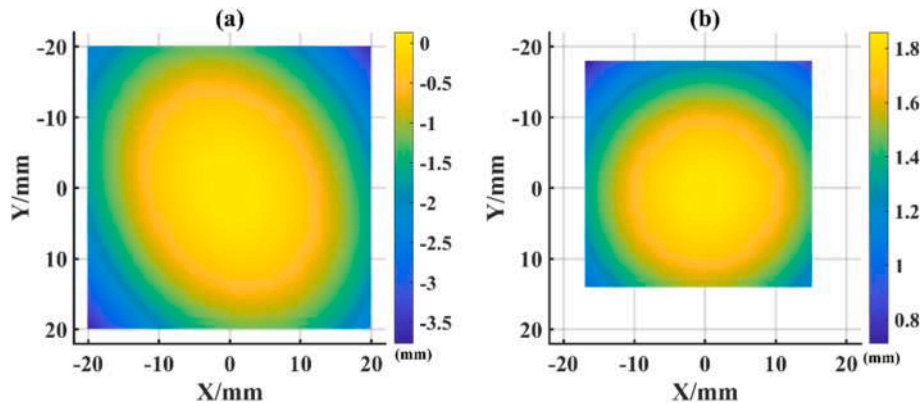


Fig. 14. The reconstruction results of the astigmatic eyeglass lens. (a) Front surface; (b) Rear surface.

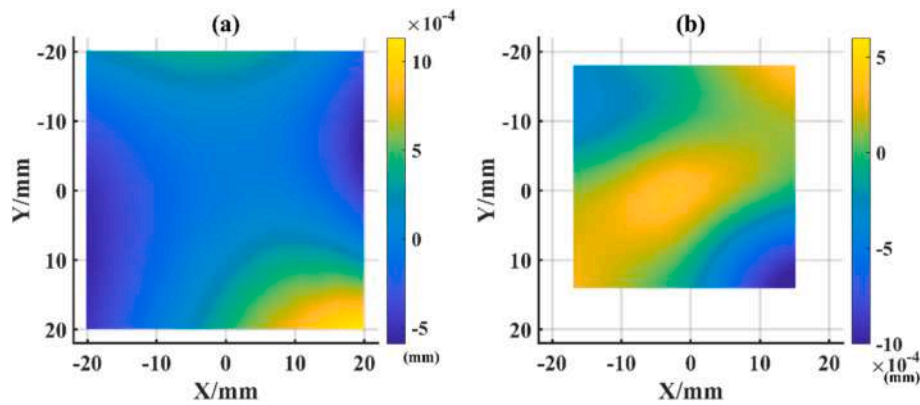


Fig. 15. The residuals of astigmatic eyeglass lens. (a) Front surface; (b) Rear surface.

Table 2

The accuracy of major and minor axes a and b of the front surface, and the spherical radius R of the rear surface. The standard values (measured by a blue light 3D scanner (ATOS Capsule 12 M)), measured values (measured by MWR-GO method), and relative deviations obtained by fitting the upper and lower surfaces with an ellipsoidal paraboloid surface and a spherical surface, respectively.

Parameters	a/mm	b/mm	R/mm
Standard	10.2891	13.4919	272.9487
Measured	10.2879	13.4951	272.7814
Relative deviation	0.016 %	0.024 %	0.061 %

better suited for handling surfaces with large curvatures. Furthermore, no pre-processing (such as powder spraying) or post-processing (such as cleaning) is required during measurement, enabling in-situ detection during industrial production. This method significantly enhances measurement speed, with the time required to measure a single element reduced to a few seconds. This high-precision, high-speed, and high-dynamic-range deflectometry system for double-layer lenses has profound significance for the field of precision freeform surface machining and inspection.

CRedit authorship contribution statement

Yiqian Gao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Haoyun Wei:** Writing – review & editing, Supervision. **Yan Li:** Writing – review & editing, Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2025.116875>.

Data availability

Data will be made available on request.

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